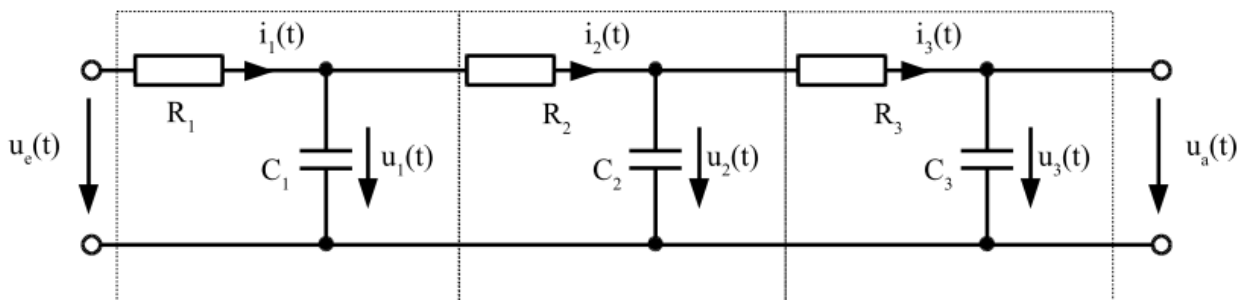


State-Space Lab 1:

State Space Representation

1 Preparation Exercises (to be completed in advance of the lab session!)

The 3-level RC network shown in the following figure is given:



- Develop the state-space model of the system.
(State variables $\underline{x} = (u_1 \ u_2 \ u_3)^T$, input variable u_e , output variable u_a)
- Create the state-space model in MATLAB with the parameters
 $R_1 = R_2 = R_3 = 4$, $C_1 = C_2 = C_3 = 1$.
- Plot the pole-zero map of the system.
Is the system stable? Is the system oscillatory? Explain your answers.
- Plot the step response of the system for a step input $u_e = 1$.
(Plot all state variables as well as the input variable in a single figure; add title, axis labels, meaningful legends.)
- Plot the unforced response of the system to an initial condition $\underline{x}_0 = (0.5, 0, 0)^T$.
(Plot all state variables in a single figure. Add title, axis labels, meaningful legends.)
- Plot the unforced response of the system to an initial condition $\underline{x}_0 = (0, 0.5, 0)^T$.
(Plot all state variables in a single figure. Add title, axis labels, meaningful legends.)
- Plot the unforced response of the system to an initial condition $\underline{x}_0 = (0, 0, 0.5)^T$.
(Plot all state variables in a single figure. Add title, axis labels, meaningful legends.)

2 Exercises for the Lab Session

Consider the 3-tank system shown in Figure 1. Three cylindrical tanks, each with base area A , are connected in series by cylindrical pipes. The connecting pipes are equipped with valves.

In addition, each tank has an outlet valve with flow coefficients k_1 , k_2 and k_3 , respectively.

The outflowing liquid (distilled water) is collected in a reservoir which supplies the pump. The flow rate q_1 of the pump is scaled to a range of -100% to 100% . All parameters and coefficients are normalized.

Pressure sensors are used for the liquid level measurements $h_1(t)$, $h_2(t)$, $h_3(t)$ in the three tanks. The measured liquid levels are scaled to a range of $0 \dots 100\%$. $h_{\max} = 100\%$ denotes the highest possible liquid level.

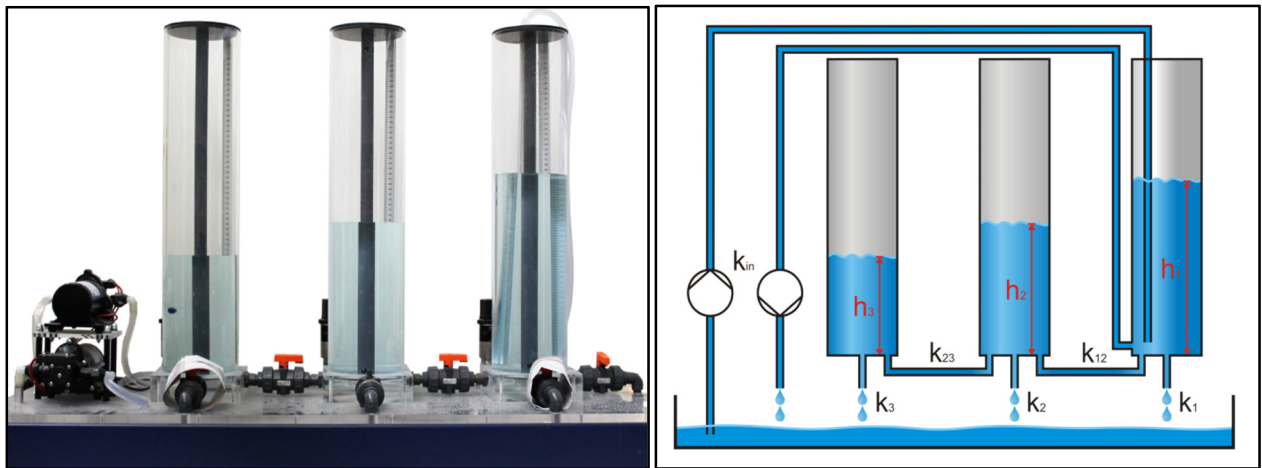


Figure 1: Photo and schematic representation of the 3-tank system

The 3-tank system can be represented by a linearized state-space model:

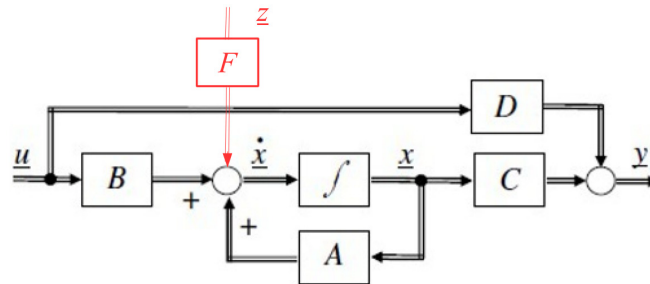
$$\dot{\underline{x}}(t) = \begin{pmatrix} -(k_1 + k_{12}) & k_{12} & 0 \\ k_{12} & -(k_{12} + k_{23} + k_2) & k_{23} \\ 0 & k_{23} & -(k_{23} + k_3) \end{pmatrix} \cdot \underline{x}(t) + \begin{pmatrix} k_{in} \\ 0 \\ 0 \end{pmatrix} \cdot u(t)$$

$$y(t) = (0 \quad 0 \quad 1) \cdot \underline{x}(t)$$

with the state vector $\underline{x} = \begin{pmatrix} h_1 \\ h_2 \\ h_3 \end{pmatrix}$.

The input variable $u = q_1$ is the flow rate of the pump. The output variable $y = h_3$ is the level of tank 3.

- Implement the state-space model of the 3-tank system in MATLAB.
Use $k_1=0.002$; $k_{12}=0.077$; $k_2=0.004$, $k_{23}=0.081$; $k_3=0.006$; $k_{in}=0.034$.
- Determine the eigenvalues of the 3-tank system.
Justify whether the system is stable. Estimate the 99% settling time of the system.
- Plot the unforced state responses of the plant to an initial condition $x_0 = (60\% \ 50\% \ 40\%)^T$.
(Plot all state variables in a single figure. Add title, axis labels, meaningful legends.)
- Plot the unforced state responses of the plant to an initial condition $x_0 = (0\% \ 0\% \ 40\%)^T$.
(Plot all state variables in a single figure. Add title, axis labels, meaningful legends.)
- Plot the three-dimensional state trajectories of the unforced responses from (c) and (d) in a single figure.
- Implement the state-space model of the 3-tank system in Simulink, including disturbances $\underline{z}(t)$ representing leaks in each of the tanks: $\dot{\underline{x}}(t) = A \cdot \underline{x}(t) + \underline{b} \cdot u(t) + F \cdot \underline{z}(t)$.



Plot the state responses of the plant to the following input $u(t)$ and disturbances $\underline{z}(t)$:

$t=0$: step change in $u(t)$ from 0% to 10%,

$t=2000$: step change in $u(t)$ from 10% to 25%,

$t=4000$: step change in $z_1(t)$ from 0% to -2% (leak in tank 1, flow coefficient $k_{z1}=0.06$),

$t=6000$: step change in $z_2(t)$ from 0% to -5% (leak in tank 2, flow coefficient $k_{z2}=0.06$).

Set the simulation time in Simulink to $t_e=10000$. Ensure that the data is sampled with sufficiently high resolution.

Plot the state responses $x_1(t)$, $x_2(t)$, $x_3(t)$ as well as $u(t)$, $z_1(t)$, $z_2(t)$ in a single figure.

Add title, axis labels, meaningful legends.

- The system is modified by adding a second input $u_2(t) = q_2(t)$, representing an inflow to tank 3, with an input gain of $k_{in2}=0.034$.
Specify the state-space model of the modified system and implement it in MATLAB.
Update your Simulink model accordingly.
- Plot the state responses of the modified plant to the following inputs and disturbances:
 $t=0$: step change in $\underline{u}(t)$ from $(0\% \ 0\%)^T$ to $(5\% \ 5\%)^T$,
 $t=2000$: step change in $\underline{u}(t)$ from $(5\% \ 5\%)^T$ to $(10\% \ 15\%)^T$,
 $t=4000$: step change in $z_1(t)$ from 0% to -2% (leak in tank 1, flow coefficient $k_{z1}=0.06$),
 $t=6000$: step change in $z_2(t)$ from 0% to -5% (leak in tank 2, flow coefficient $k_{z2}=0.06$).

Plot the state responses $x_1(t)$, $x_2(t)$, $x_3(t)$ as well as $u(t)$, $z_1(t)$, $z_2(t)$ in a single figure.

Add title, axis labels, meaningful legends.